

Unit 14: Data interpretation

CONTENTS

Objective

Introduction

14.1 Bar Diagram

14.2 Cubes and Dices

14.3 Dices:-

14.4 A table

14.5 Cumulative Bar Graph

14.6 Pie Charts

Summary

Keywords

Self Assessment

Answers for Self Assessment

Review Questions

Further Readings

Objective

The key concepts explained in this unit, including area a learner is expected to learn and master after going through the unit are: -

- We will understand simplest way to represent data.
- We will understand how to show a particular data line graph
- We will understand how to visualize the presentation of categorical with bar diagram.
- We will understand how to use table to represent data.
- We will understand how to use pie chart to represent data.

Introduction

A line graph (or line chart) is a graph which displays information as a series of data points connected by straight line segments. A line graph shows a particular data that change at equal intervals of time.

Line graph is the simplest way to represent data. Single set or multiple sets of data can be shown in a graph. Figure 1 shows a line graph representing average daily temperature for two states, A and B.

Usually, we can analyze the following things using line graphs:

1. Increase in profit in absolute terms or in percentage terms
2. Average annual growth rate.
3. Average profit
4. Capability utilization.

$$\begin{aligned} \text{Capacity utilization} &= \\ \text{capacity utilization} &= \frac{\text{total production}}{\text{total capacity}} \times 100 \end{aligned}$$

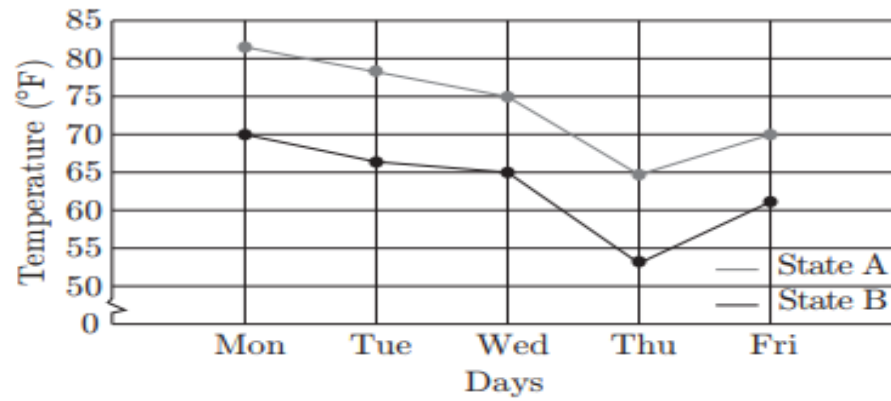


Figure 1 | Line graph representing average daily temperature for two states.



Example 1:- The total capacity of a Hyundai i10 plant is 150 cars per day. In the month of April 2009, the plant manufactured at the rate of 120 cars per day. Find the capacity utilization in the month of April.

Solution : We know that capacity utilization is given by

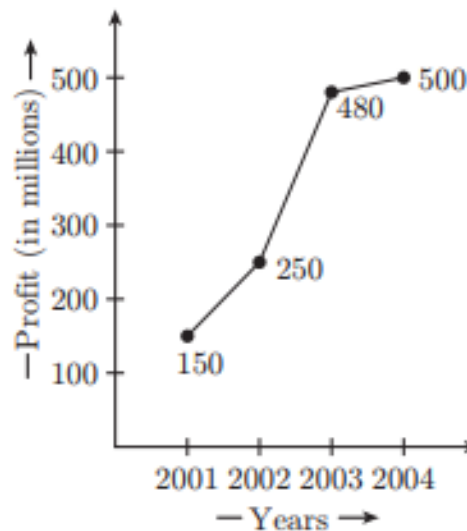
$$\text{Capacity utilization} = \frac{\text{Total production}}{\text{total capacity}} \times 100$$

$$= \frac{120}{150} \times 100$$

$$= 80\%$$



Example 2: Direction for Q2 and Q3: Consider the following graph:



Balance sheet of XYZ Corporation

2. What was the percent increase in profit in 2001 – 02 and 2003 – 04?

Solution: Percent increase in profit in 2001 – 2002

$$= \frac{250 - 150}{150} \times 100$$

$$= 66.67\%$$

Present increase in profit in 2003-04

$$= \frac{500 - 480}{480} \times 100$$

$$= 4.16\%$$

3. What was the average annual growth rate for XYZ Corporation?

Solution: Average annual growth rate

$$= \frac{\text{increase in profit for duration}}{\text{Base years profit}} \times \frac{100}{\text{number of years}}$$

$$= \frac{500 - 150}{150} \times \frac{100}{3}$$

$$= \frac{350}{150} \times \frac{100}{3}$$

$$= 66.67\%$$



Example 3:- Direction for Q4 and Q5: Consider the following graph

The above graph is showing consumption of metals versus plastics in the given years (2006 – 11) for car manufacturing.

4. Which item (and for which year) shows the highest percentage change in consumption over the previous year?

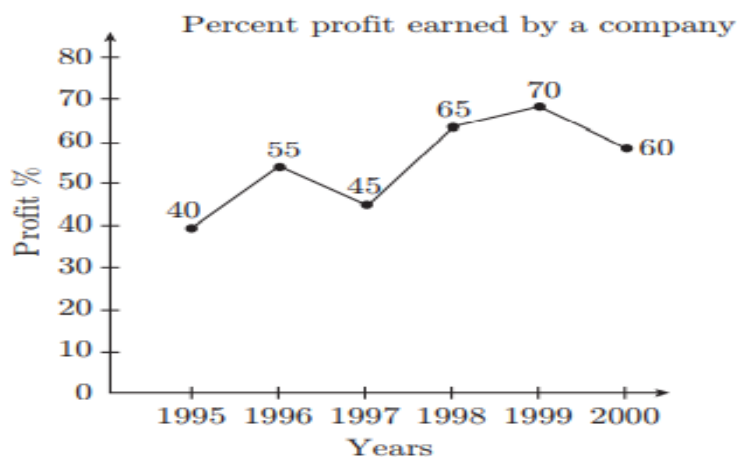
Solution: Percentage change in consumption is highest for metals in 2008. 5.

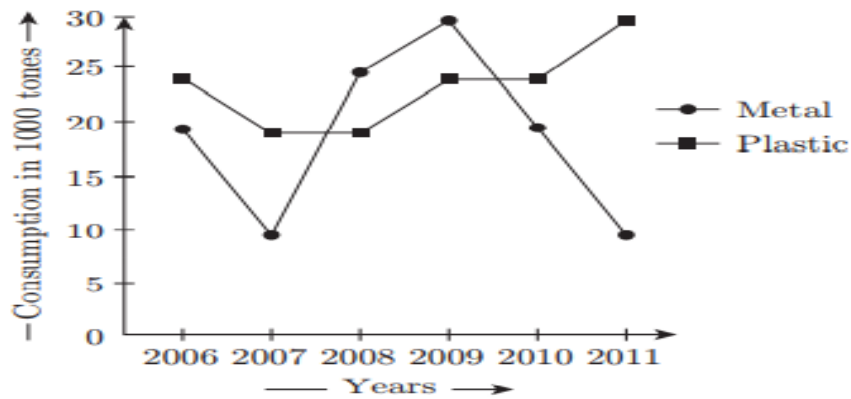
5. Find out the number of years for which the consumption of metals was less than the consumption of plastic over the given time period.

Solution: For 2006, 2007, 2010 and 2011, the consumption of metals was less than plastics. Hence, for four years the consumption of metals was less than plastics.



Example 4:- Direction for Q6 – Q8: Consider the following graph:





6. If the expenditures in 1996 and 1999 are equal, then what is the approximate ratio of the income in 1996 and 1999, respectively? Solution: Let the expenditure in 1996 be x .

Also, let the incomes in 1996 and 1999 be a and b , respectively.

Then, for the year 1996, we have

$$\frac{a-x}{x} \times 100$$

$$55$$

$$a = \frac{155x}{100}$$

$$\frac{b-x}{x} \times 100 = 70$$

$$b = \frac{170x}{100}$$

From equ (1) and (2) we get

$$\frac{a}{b} = \frac{(155x/100)}{(170x/100)}$$

$$= \frac{155}{170} \approx 9$$

7. What is the average profit earned for the given years?

Solution: Average profit for the given years

$$= \frac{40+55+45+65+70+60}{6}$$

$$= \frac{335}{6}$$

$$= 55.83$$

8. If the income in 1998 was ₹264 crores, what was the expenditure in 1998?

Solution : Let the expenditure in 1998 be ₹ x crores.

$$\frac{264-x}{x} \times 100$$

$$= 65$$

$$X = \frac{264x \cdot 100}{165}$$

$$160$$

Expenditure in 1998 = ₹ 160 crores.

14.1 Bar Diagram

A bar diagram or bar graph is a chart with rectangular bars in which the lengths of bars are in proportion to the values of the entities they represent.

A vertical bar chart is called a column bar chart (Fig. 1a) and a horizontal bar chart is called a row bar chart (Fig. 1b).

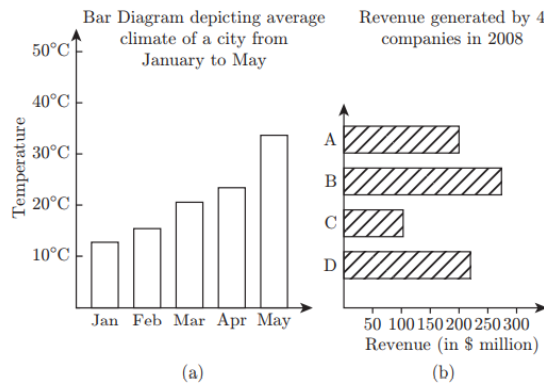


Figure 1 | Bar diagram: (a) Column bar chart and (b) row bar chart.

Bar diagrams are often used to provide visual presentation of categorical data and are considered very reliable. Bar graphs can also be used for comparison of complex data. Also, a bar chart is very useful for displaying discrete data values.

Multiple sets of interrelated data can be represented using a multiple bar diagram. It is used to compare more than one quantity. The technique of a simple bar diagram is used to draw the diagram. However, different quantities are represented using different shapes, sizes, colours, patterns, etc. Figure 2 shows a regular multiple bar diagram.

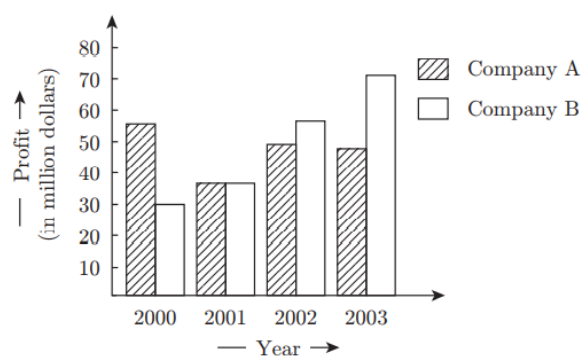


Figure 2 | Multiple bar diagram.

A compound bar diagram is used to combine or compare two or more types of information in one chart. The bars are stacked on top of one another. However, different quantities are represented using different shapes, sizes, colours, patterns, etc. Figure 3 shows a regular compound bar diagram.

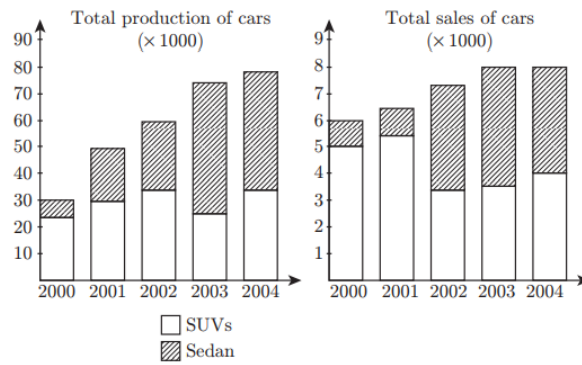


Figure 3 | Compound bar diagram.

A bar chart may be drawn on a percentage basis. To obtain a percentage bar diagram, bars of length equal to 100 for each class are drawn and subdivided in the proportion of the percentage of their component. Figure 4 shows a regular percentage bar diagram

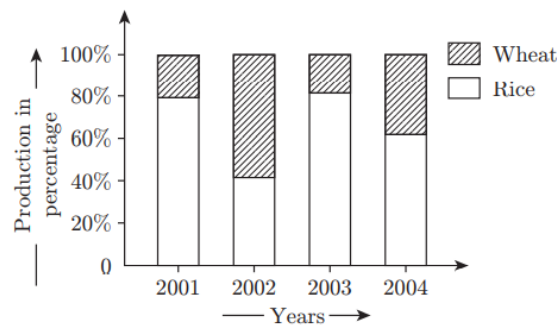
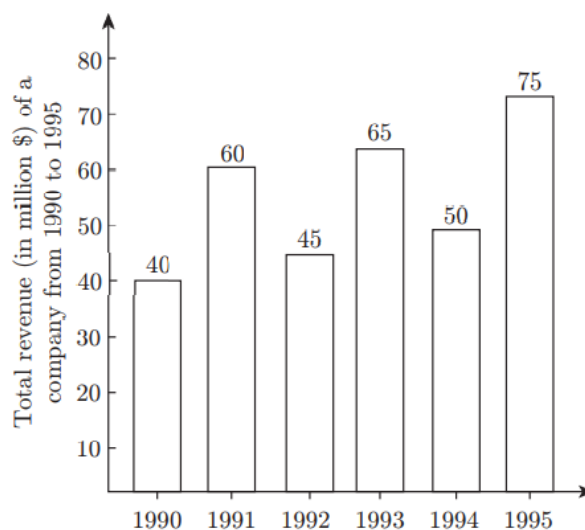


Figure 4 | Percentage bar diagram.

Example 5:- Direction for Q1-Q4: Study the following bar diagram carefully and answer the questions.



Unit 14: Data interpretation

Which pair of years had the same average revenue as that of 1990 and 1995? Solution: Total revenue in 1990 = \$ 40 million Total revenue in 1995 = \$ 75 million

$$\text{Average revenue in 1990 and 1995} = \frac{40+75}{2} = \$ 57.5 \text{ million}$$

Total revenue in 1993 = \$ 65 million

$$\text{Total revenue in 1994} = \$ 50 \text{ million Average revenue for 1993 and 1994} = \frac{65+50}{2} = \$ 57.5 \text{ million}$$

Hence, average revenue of 1993 and 1994 was same as 1990 and 1995.

2. In which year(s), the percentage increase in production was maximum from the previous year?

Solution: Percentage increase from 1990 to 1991 is

$$\frac{60-40}{40} \times 100$$

$$\frac{20}{40} \times 100$$

$$= 50\%$$

Also, percentage increase from 1994 to 1995 is

$$\frac{75-50}{50} \times 100 = 50\%$$

Thus, percentage increase was maximum in 1991 and 1995.

3. . What was the percentage drop in the revenue from 1991 to 1992?

Solution: Total revenue in 1991 = \$ 60 million Total revenue in 1992 = \$ 45 million

Percentage drop was

$$\frac{60-45}{60} \times 100$$

$$= \frac{15}{60} \times 100$$

$$= 25\%$$

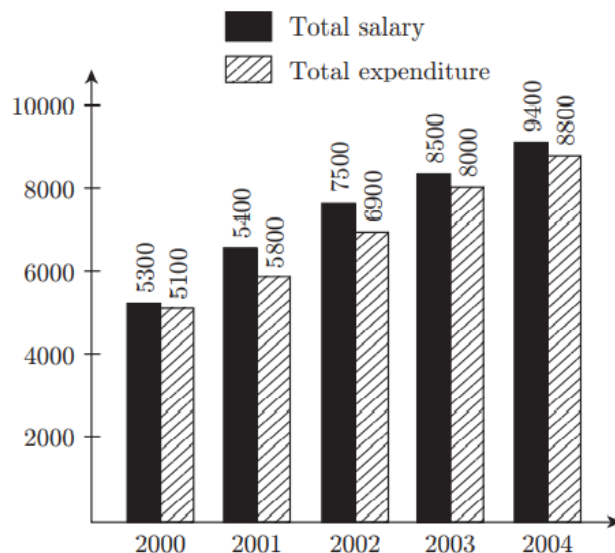
4. What was the difference between revenue in 1995 and 1990?

Solution: Total revenue in 1995 = \$ 75 million Total revenue in 1990 = \$ 40 million

Difference in revenue = 75 - 40 = \$ 30 million



Example 6:- Direction for Q5-Q7: Study the following bar diagram carefully and answer the questions.



Analytical Skills-I

5. What was the percentage increase in the "Total Salary" in 2002 as compared to 2000?

Solution: Total salary in 2002 = 7500

Total salary in 2000 = 5300

Percentage increase was $\frac{7500-5300}{7500} \times 100 = 5.88\%$

6. If profit = total salary - expenditure, then in 2003 what percentage of total salary was the profit made?

Solution: Total salary in 2003 = 8500

Total expenditure in 2003 = 8000

Total profit = 500

Profit % = $\frac{500}{8500} \times 100 = 5.88\%$

7. By what amount had the expenditure increased over the period 2000-2004?

Solution: Total expenditure in 2000 = ₹5100

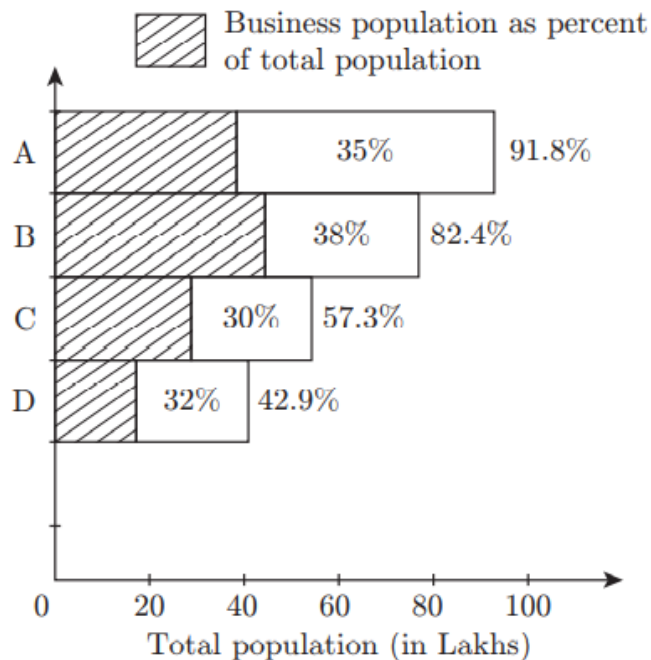
Total expenditure in 2004 = ₹8800

Total increase in expenditure from 2000-2004 was

8800 - 5100 = ₹3700



Example 7:- Direction for Q8-Q10: Study the following bar diagram carefully and answer the questions.



8. What is total business population of A?

Solution: Total population of A = 91.8 lakh

Total business population = 35%

$\frac{35}{100} \times 9.18 \text{ lakh} = 32.13 \text{ lakh}$

9. What place has the highest business population?

Solution: Business population of A = $\frac{35}{100} \times 9.18 \text{ lakh} = 32.13 \text{ lakh}$

Business population of B = $\frac{38}{100} \times 8.24 \text{ lakh} = 31.31 \text{ lakh}$

Business population of C = $\frac{30}{100} \times 57.3 \text{ lakh} = 17.19 \text{ lakh}$

Business population of D = $\frac{32}{100} \times 42.9 \text{ lakh} = 13.728 \text{ lakh}$

Hence, A has the highest business population.

10. What is the difference between business populations of C and D?

Solution: Business population of C = $\frac{30}{100} \times 57.3 \text{ lakh} = 17.19 \text{ lakh}$

Business population of D = $\frac{32}{100} \times 49.9 \text{ lakh} = 13.728 \text{ lakh}$

Difference between business population of C and D = $17.19 - 13.728 = 3.462 \text{ lakh}$

14.2 Cubes and Dices

A cuboid is any three-dimensional figure with length, breadth and height. If all the three sides of a cuboid are same, then the figure is called a cube. Figure 1 shows a Rubik's cube which is a cube made of 27 smaller cubes.

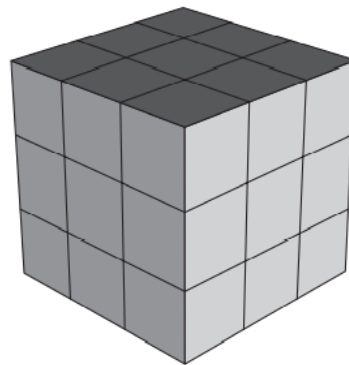


Figure 1 | Rubik's cube

In a cube:

1. There are 8 corners/vertices.
2. There are 6 faces, all equal in area.
3. There are 12 edges, all equal in area.
4. It should be noted that

Number of edges Number of vertices Number of face = + - 2

Some important tips to solve questions based on cubes are as follows:

If we paint a cube of the dimension $n \times n \times n$ in any one colour and cut it to get n^3 cubelets, then the number of cubes with only one face painted = $(n-2)^2 \times 6$

2. If we paint a cube of the dimension $n \times n \times n$ in any one colour and cut it to get n^3 cubelets, then the number of cubes with two faces painted = $(n-2)^2 \times 12$

3. If we paint a cube of the dimensions $n \times n \times n$ in any one colour and cut it to get n^3 cubelets, then the number of cubes with three faces painted = $(n-2)^3$

14.3 Dices:-

Dices are small cubical structures with numbers 1 to 6 (or dots) marked on each face.

When sum of number of opposite pair of faces is same, then the dice is called a symmetric dice. Hence, for a symmetrical dice, the faces opposite to each other will be $1 \rightarrow 6$, $2 \rightarrow 5$, $3 \rightarrow 4$. Figure 2 shows symmetrical dices.

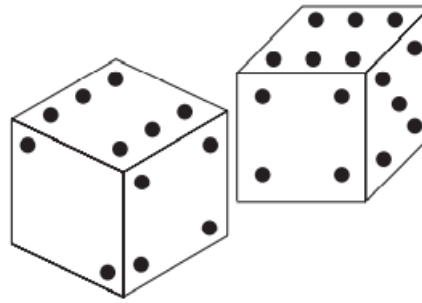


Figure 2 | Symmetric dices.

When sum of numbers on opposite pair of faces is different, then the dice is called an asymmetric dice. Figure 3 shows asymmetrical dices.

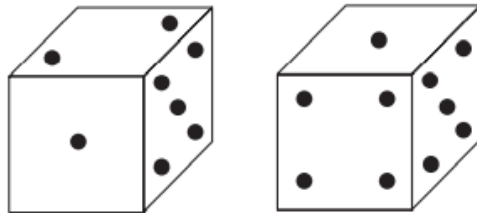


Figure 3 | Asymmetric dices.

Solved Example :-



Example 8: Direction for Q1–Q5: If a cube is cut into n^3 identical cubelets using minimum number of cuts, after painting all faces of cubes with white colour, then answer the following question

1. What is the maximum number of cuts required?
Solution: There are a minimum of $(n - 1)$ equidistant cuts parallel to each of three faces which are joining the corner. Hence, total number of cuts required is $3(n - 1)$.
2. How many cubelets will have exactly two faces painted?
Solution: Cubes with exactly two faces painted = $12(n - 2)$, where n is the number of parts into which side is divided.
3. How many cubelets will have exactly one face painted?
Solution: Cubes with exactly one face painted = $6(n - 2)^2$, where n is the number of parts into which side is divided.
4. How many cubelets will have at most two faces painted?
Solution: Total number of cubelets with at most two faces painted = Total number of cubes – Number of cubes with three faces painted = Total number of cubes = n^3
Number of cubes with three faces painted = 8

Hence, total number of cubelets with at most two faces painted = $n^3 - 8$
5. How many cubelets will have at least one face painted?
Solution: Total number of cubelets with at least one face painted = Total number of cubes – Number of cubes with no-face painted.
Number of cubes with no face painted = $(n - 2)^3$
Hence, total number of cubelets with at least one face painted = $n^3 - (n - 2)^3$

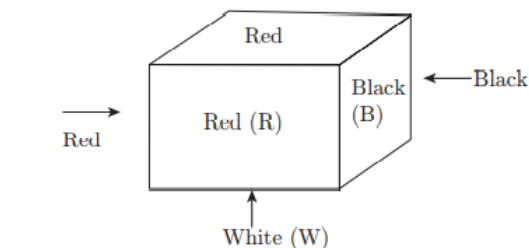
Three adjacent faces of one cube are painted in red, one adjacent pair of faces is painted in black and remaining faces are painted in white.



The cube is then cut into 216 identical cubelets.

Example 9: Direction for Q6–Q10: A cube is cut into 216 cubelets = $6 \times 6 \times 6$

We can draw the pattern of painting as shown in the following figure:



Now, let us analyze the colour combination.

Corners:

$$RRR = 1, BBR = 1$$

$$RRB = 2, BBW = 1$$

$$RRW = 1, RWB = 2$$

Edges:

$$RR = 3, RW = 2$$

$$RB = 3, BW = 2$$

$$BB = 1$$

Faces:

$$R = 3 \quad B = 2 \quad W = 1$$

6. How many cubelets have all the three colour on them?

Solution: The small pieces will be formed from the corners. The cubelets having all three colours (red, black and white) are 2.

7. How many cubelets have exactly two colour on them?

Solution: The cubelets with exactly two colour found at corners = 5

Total edges having different colour on either side = 8

Therefore, total cubelets = $8(6-2)+5 = 37$

8. How many cubelets have exactly one colour on them?

Solution: The cubelets having exactly one colour at corner (RRR) = 1

At the three edges each of RR categories = $3(6-2) = 12$

At one edge of BB category = $1(6-2) = 4$

At middle of each of the six faces = $6(6-2) = 96$

Hence, total cubelets with exactly one colour = $1 + 12 + 4 + 96 = 113$

9. How many cubelets do not have red colour on them?

Analytical Skills-I

Solution: The number of cubelets with no red colour = Total cubes – (Total cubelets from one red surface + Total cubelets from two red surface + Total cubelets from three red surfaces) = $216 - (36 + 30 + 25) = 125$

10. How many cubelets have black or white colour on them but not red colour on them?

Solution: The number of cubelets with no red colour = 125

The number of cubelets with black or white but not red = $125 - (6 - 2)^3 = 125 - 64 = 61$

14.4 A table

A table is used to represent data in rows and columns.

Tables are arguably the most common way of arranging a given data. Evidently, tables are used almost everywhere, in print media, books, presentation slides, etc.

Tables differ significantly in variety, structure, flexibility, notation, representation and use.

Tables are advantageous to us as they can arrange a large amount of data easily and without any complexity.

However, locating pattern or visual observations are difficult.



Example 10: Direction for Q1 – Q5: The following table shows sales of cars of five companies over the years (in thousand).

Company	Years				
	2000	2001	2002	2003	2004
Honda	20	21	50	35	75
Hyundai	29	31	23	46	42
Maruti	31	29	27	22	16
Tata	33	14	33	37	48
Skoda	15	17	32	39	47
Total	128	112	165	179	228

Consider the table and answer the following questions.

- For which company did the amount of sales of cars increase continuously over the years?
Solution: For Skoda, we can see that the sales of each year are always greater than the previous year.
- For which company did the amount of sales of cars decrease continuously over the years?
Solution: For Maruti, we can see that the sales of each year are always lesser than the previous year.
- In 2001, which company had the biggest increase in sales from 2000?

Solution: Three companies had more sales in 2001 than 2000.

$$\text{Percentage increase for Honda} = \frac{21-20}{20} \times 100 = 5\%$$

$$\text{Percentage increase for Hyundai} = \frac{31-29}{29} \times 100 = 6.9\%$$

$$\text{Percentage increase for Skoda} = \frac{17-15}{15} \times 100 = 13.33\%$$

Hence, Skoda had the biggest percentage increase in sales of 2001.

- In which year did Maruti have the largest percentage decrease in sales? Solution:

$$\text{Percentage decrease in sales in 2001} = \frac{31-29}{20} \times 100 = 6.45\%$$

$$\text{Percentage decrease in sales in 2002} = \frac{29-27}{27} \times 100 = 18.52\%$$

$$\text{Percentage decrease in sales in 2003} = \frac{22-16}{16} \times 100 = 18.52\%$$

$$\text{Percentage decrease in sales in 2004} = \frac{22-16}{16} \times 100 = 27.27\%$$

Hence, in 2004, the decrease in sales for Maruti was the highest

5. Which company sold the most number of cars in the five years?

Solution: Total sales of Honda = $20 + 21 + 50 + 35 + 75 = 201000$

Total sales of Hyundai = $29 + 31 + 23 + 46 + 42 = 171000$

Total sales of Maruti = $31 + 29 + 27 + 22 + 16 = 125000$

Total sales of Tata = $33 + 14 + 33 + 37 + 48 = 165000$

Total sales of Skoda = $15 + 17 + 32 + 39 + 47 = 150000$

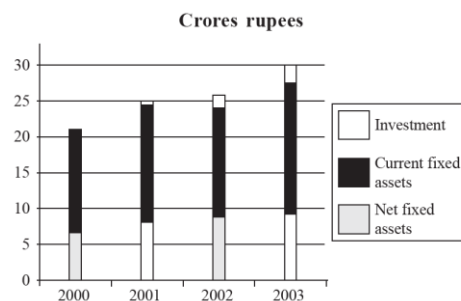
Hence, Honda had the highest sales from 2000 to 2004.



Example 11: Direction for Q6–Q10: The following table shows total revenue of five companies over the years (in \$ millions).

14.5 Cumulative Bar Graph

In a cumulative bar graph, the length of the bar is divided proportionately among various quantities represented in the graph. Thus, it may be conveniently used for making comparisons, Example is as follows



14.6 Pie Charts

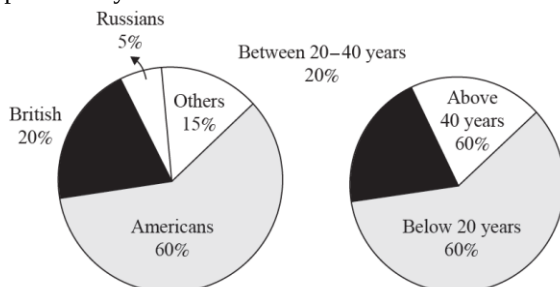
In a pie chart, the given data is distributed over a circle. Each part of the data makes a certain central angle.

For example, if from all the questions asked in a Bank PO exam, 25% are on data interpretation, then central angle made by this term

$$= (25/100) \times 360^\circ = 90^\circ$$

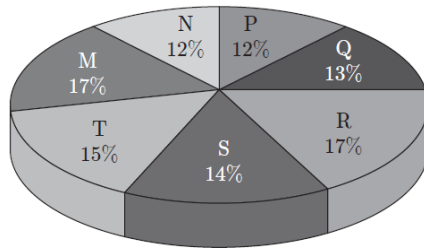
Pie charts are useful for representing percentages or proportions of various elements with respect to the total quantity. They also represent shares of various parts of a particular quantity.

Example : The following pie charts describe the characteristics of foreign tourists visiting India in a particular year.

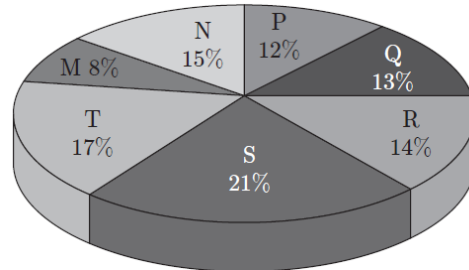


Distribution of students at graduate and post graduate levels in seven institutes (M, N, P, Q, R, S and T).

Total number of students of graduate level = 27300.

Analytical Skills-I

Total number of students of post graduate level = 24700.



1. How many students of institutes of M and S are studying at graduate level?

Solution: Students of institute M at graduate level =

$$(17/100) \times 27300 = 4641$$

Students of institute S at graduate level =

$$(14/100) \times 27300 = 3822$$

$$= 14/100 \times 27300 = 3822$$

Total number of students at graduate level in institutes

$$\text{M and S} = 4641 + 3822 = 8463.$$

2. What is the total number of students studying at postgraduate level from institutes N and P?

Solution: Students studying at postgraduate level from N = $(15/100) \times 24700 = 3705$.

Students studying at postgraduate level from P = $(12/100) \times 24700 = 2964$.

Required number = $3705 + 2964 = 6669$.

3. What is the ratio between the number of students studying at postgraduate and graduate levels from institute S?

Solution: Number of postgraduate students from institute S = $(21/100) \times 24700 = 5187$

Number of graduate students from institute S = $(14/100) \times 27300 = 3822$

Required ratio = $(5187/3822) = 19/14$.

Summary

The key concepts learned from this unit are: -

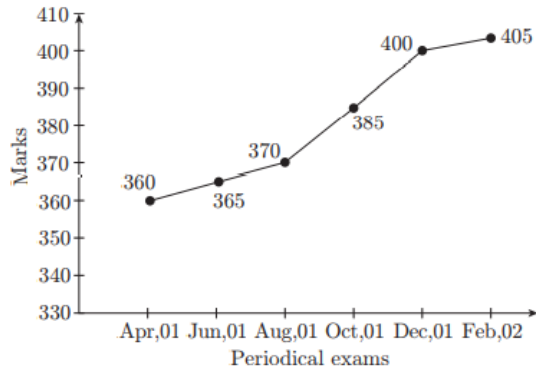
- We have learnt how to We will understand how to show a particular data line graph
- We have learnt how to visualize the presentation of categorical with bar diagram.
- We have learnt how to use table to represent data.
- We have learnt how to use pie chart to represent data.

Keywords

- Bar Diagram
- Line graph
- Dices
- Pie Diagram

Self Assessment

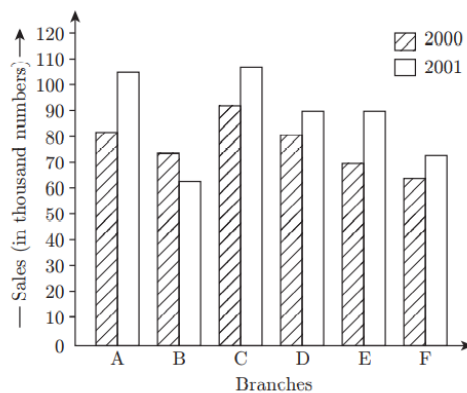
Direction for Q1 and Q3: Consider the following graph:



The above graph shows marks obtained by a student in six periodicals held in every two months during the year in the session 2001 – 2002. In addition, the maximum total marks in each periodical exam are 50

- What is the percentage of marks obtained by the student in the periodical exams of August, 01 and October, 01 taken together?
 - 75.5%
 - 72%
 - 30%
 - 45%
- What are the average marks obtained by the student in all the periodical exams during the last session?
 - 380.83
 - 340
 - 370
 - 289

Direction for Q3-Q7: Study the following bar diagram carefully and answer the questions.



- What is the ratio of the total sales of branch B for both years to the total sales of branch D for both years?
 - Required ratio = $\frac{140}{180} = \frac{7}{9}$
 - Required ratio = $\frac{140}{180} = \frac{7}{10}$

Analytical Skills-I

C. Required ratio = $\frac{143}{180} = \frac{7}{9}$

D. Required ratio = $\frac{140}{190} = \frac{7}{9}$

4. What is the total sale of branches A, C and E together for both the years (in thousands)?

Solution: Total sales of branches A, C and E for both the years (in thousands)

A. 660

B. 560

C. 760

D. 860

5. What percent of the average sales of branches A, B and C in 2001 is the average sales of branches A, C and F in 2000?

A. 77.5%

B. 87.5%

C. 88.5%

D. 98.5%

6. Total sale of branch F for both the years is what percent of the total sales of branch C for both the years?

A. 70.5%

B. 12%

C. 72%

D. 73.17%

7. What is the total number of red faces?

A. 120

B. 720

C. 620

D. 320

9. How many cubes have two adjacent blue faces each?

A. 2 cubes

B. 62 cubes

C. 64 cubes

D. 60 cubes

10. How many cubes have only one red face each?

A. only one face red.

B. only two face red.

C. only three face red.

D. only four face red.

11. Which two colors have the same number of faces?

- A. First 64 cubes are such that each of whose two faces are green and second 64 cubes are such that each of whose two faces are blue. Therefore, green and blue colour have the same number of faces.
- B. First 64 cubes are such that each of whose two faces are red and second 64 cubes are such that each of whose two faces are blue. Therefore, red and blue colour have the same number of faces.
- C. First 64 cubes are such that each of whose two faces are green and second 64 cubes are such that each of whose two faces are red. Therefore, green and red colour have the same number of faces.
- D. First 46 cubes are such that each of whose two faces are green and second 64 cubes are such that each of whose two faces are blue. Therefore, green and blue colour have the same number of faces.

Consider the table and answer the following questions.

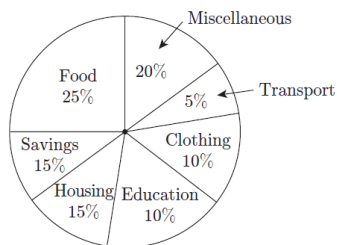
Company	Years			
	2000	2001	2002	2003
A	20	17	8	18
B	10	12	16	20
C	17	15	4	2
D	6	16	3	9
E	18	30	10	11

12. What was the total revenue of company D from 2000 to 2003?
- A. \$ 34 million
B. \$ 44 million
C. \$ 54 million
D. \$ 64 million
13. What was the revenue of all the companies in the year 2002?
- A. \$ 11 million
B. \$ 21 million
C. \$ 31 million
D. \$ 41 million
14. From 2000 to 2003, in which company was there a continuous increase in revenue?
- A. Company A had a continuous increase in revenue from 2000 to 2003.
B. Company B had a continuous increase in revenue from 2000 to 2003.
C. Company C had a continuous increase in revenue from 2000 to 2003.
D. Company D had a continuous increase in revenue from 2000 to 2003.
15. In which two years did company B and D have the same amount of revenue?
- A. in 1999 and 2001.
B. in 2001 and 2002
C. in 2004 and 2005
D. in 2000 and 2001
16. What was the percentage decrease in total revenue from 2001 to 2003?

Analytical Skills-I

- A. 33.3%
- B. 53.3%
- C. 63.3%
- D. 73.3%

Direction for Q9–Q12: The following pie chart shows the percentage distribution of money spent by a family in August, 2013.



Consider the figure and answer the following questions.

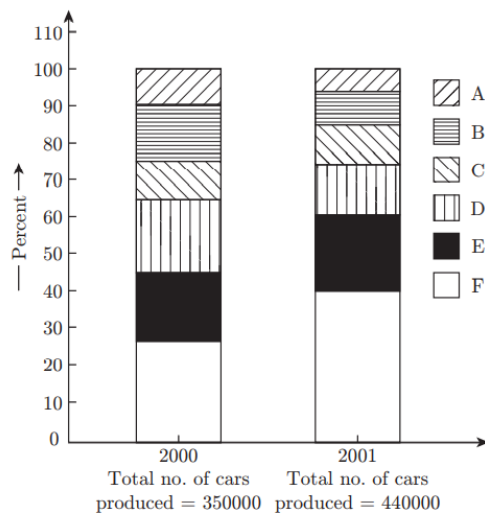
17. If the total amount spent was Rs 120000, then what was the total amount spent on "housing"?
 - A. Rs 18000
 - B. Rs 21000
 - C. Rs 15000
 - D. Rs 24500
18. What is the ratio of the total amount of money spent on clothing to that on food?
 - A. 5 : 2
 - B. 2 : 5
 - C. 3 : 4
 - D. 4 : 3
19. How much of the total money was used for savings?
 - A. 36°
 - B. 48°
 - C. 52°
 - D. 54°

Answers for Self Assessment

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. A | 2. A | 3. A | 4. B | 5. B |
| 6. D | 7. D | 8. C | 9. A | 10. A |
| 11. A | 12. D | 13. B | 14. D | 15. A |
| 16. A | 17. B | 18. D | 19. B | |

Review Questions

1. Direction for Q1-Q5: Study the following bar diagram carefully and answer the questions.



1. What is the difference in the number of E-type cars produced in 2000 and that produced in 2001?

Solution: Total number of E-type cars produced in 2001

$$= \frac{(60-40) \times 440000}{100} = 880000$$

$$\text{Total number of E-type cars produced in 2000} = \frac{(45-25) \times 440000}{100} = 880000$$

$$\text{Required difference} = 88000 - 77000 = 11000$$

2. If the percentage of production of F-type cars in 2001 were the same as that in 2000, then the number of F-type cars produced in 2001 would have been?

Solution: We are given that the percentage production of F-type cars in 2001 = percentage production of F-type cars in 2000 = 25%

$$\text{Then, number of F-type cars produced in 2001} = 25\% \text{ of } 4,40,000 = 110000$$

3. Total number of cars of model B, E and F manufactured in 2000 is?

Solution: In 2000, total number of cars produced = 350000

$$\text{a) } F = (25 - 0)\% \text{ of } 350000 = \frac{25}{100} \times 350000 = 875000$$

$$\text{b) } E = (45 - 25)\% \text{ of } 350000 = \frac{20}{100} \times 350000 = 70000$$

$$\text{c) } D = (65 - 45)\% \text{ of } 350000 = \frac{20}{100} \times 350000 = 70000$$

$$\text{d) } C = (75 - 65)\% \text{ of } 350000 = \frac{10}{100} \times 350000 = 35000$$

$$\text{e) } B = (90 - 45)\% \text{ of } 350000 = \frac{15}{100} \times 350000 = 52500$$

$$\text{f) } A = (100 - 90)\% \text{ of } 350000 = \frac{10}{100} \times 350000 = 35000$$

In 2001, total number of cars produced = 44000

$$F = (40-0)\% \text{ of } 440000 = \frac{40}{100} \times 440000 = 176000$$

$$E = (60-40)\% \text{ of } 440000 = \frac{20}{100} \times 440000 = 88000$$

$$D = (75-60)\% \text{ of } 440000 = \frac{15}{100} \times 440000 = 66000$$

$$C = (85-75)\% \text{ of } 440000 = \frac{10}{100} \times 440000 = 44000$$

Analytical Skills-I

$$B = (95-85)\% \text{ of } 440000 = \frac{10}{100} \times 440000 = 44000$$

$$A = (100-95)\% \text{ of } 440000 = \frac{5}{100} \times 440000 = 22000$$

Total number of cars of model B, E and F manufactured in 2000 = $(52500 + 70000 + 87500) = 210000$

4. If 85% of the C-type cars produced in each year were sold by the company, how many C-type cars remain unsold?

Solution: Number of C-type cars which remain unsold in 2000 = $\frac{15}{100} \times 35000 = 5250$

Number of C-type cars which remained unsold in 2001 = $\frac{15}{100} \times 44000 = 6600$

Therefore, total number of C-type cars which

remained unsold = $5250 + 6600 = 11850$

5. For which model was the percentage rise/fall in production from 2000 to 2001 minimum?

Solution: The percentage change in production from 2000 to 2001 for various models is:

$$F = \left[\frac{(176000-87500)}{105000} \times 100 \right] = 101.14\% \text{ rise}$$

$$E = \left[\frac{(88000-70000)}{70000} \times 100 \right] = 25.71\% \text{ rise}$$

$$D = \left[\frac{(70000-66000)}{105000} \times 100 \right] = 5.71\% \text{ fall}$$

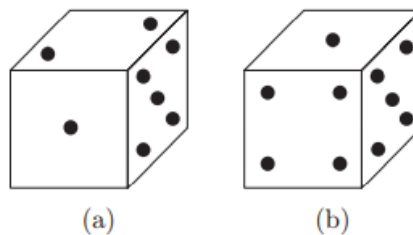
$$C = \left[\frac{(44000-35000)}{35000} \times 100 \right] = 25.14\% \text{ rise}$$

$$B = \left[\frac{(52000-44000)}{105000} \times 100 \right] = 16.19\% \text{ fall}$$

$$A = \left[\frac{(35000-22000)}{35000} \times 100 \right] = 37.14\% \text{ fall}$$

Therefore, minimum change is observed in model D.

6. Observe the dots on a dice (1–6 dots) in the following figures. How many dots are contained on the face opposite to that containing 4 dots?

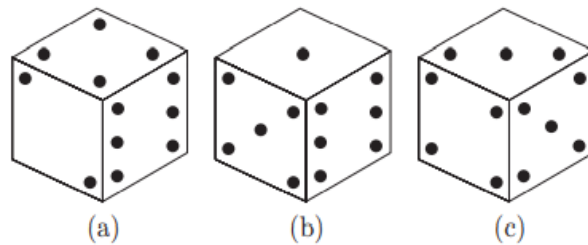


Solution: We can see from Figs. (a) and (b) that the side of the dice with 5 dots remains stationary.

Hence, when we shift the dice such that we move the side with 1 dot, we observe that one side has 2 dots and the other side has 4 dots.

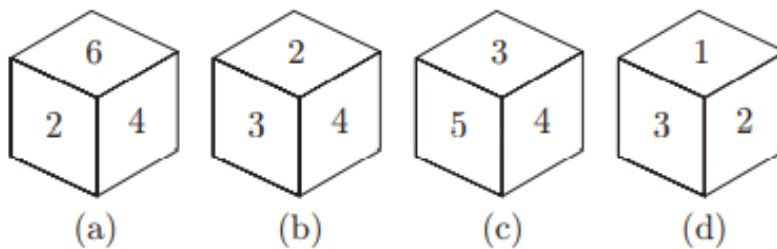
Since these two faces are opposite to each other, 2 dots are contained on the face opposite to that containing 4 dots.

7. Three different positions of a dice are shown in the following figures. How many dots lie opposite 2 dots?



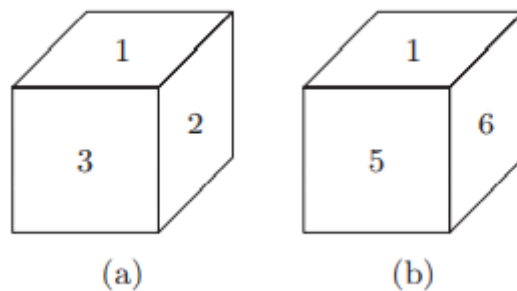
Solution: From Figs. (b) and (c), we conclude that 1, 6, 3 and 4 dots lie adjacent to 5 dots. Therefore, 2 dots must lie opposite 5 dots. Conversely, 5 dots must lie opposite 2 dots.

8. A dice is thrown four times and its four different positions are shown in the following figures. What is the number on the face opposite the face showing 2?



Solution: From Figs. (a), (b) and (d), we conclude that 6, 4, 3 and 1 lie adjacent to 2. Hence, 5 must lie opposite 2

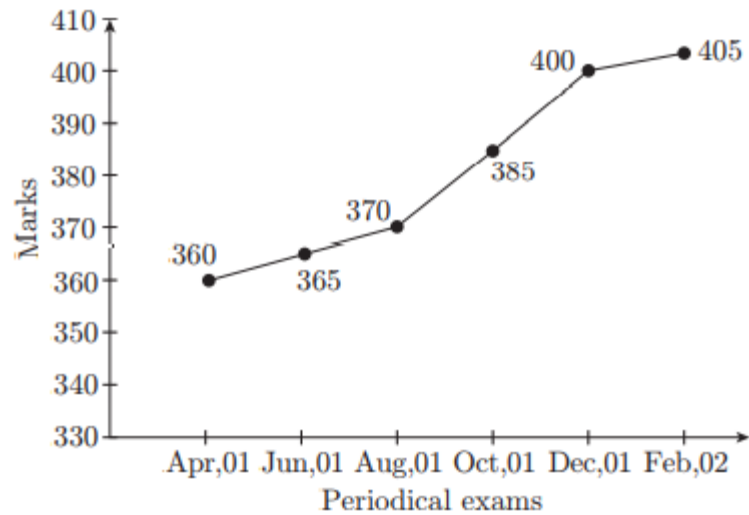
9. Two positions of a cube with its surfaces numbered are shown in the following figures. When the surface 4 touches the bottom, which surface will be on the top?



Solution: In these 2 positions, one common face with number 1 is in the same position. Also, we can see that 2, 3, 5 and 6 are adjacent to 1. Thus, 1 and 4 are opposite.

Therefore, when surface 4 touches the bottom, 1 will be on the top.

10. Consider the following graph:



The above graph shows marks obtained by a student in six periodicals held in every two months during the year in the session 2001 – 2002. In addition, the maximum total marks in each periodical exam are 50

What is the percentage of marks obtained by the student in the periodical exams of August, 01 and October, 01 taken together?



Further Readings

1. Quantitative Aptitude For Competitive Examinations By Dr. R S Aggarwal, S Chand Publishing
2. A Modern Approach To Verbal & Non-Verbal Reasoning By Dr. R S Aggarwal, S Chand Publishing
3. Magical Book On Quicker Maths By M Tyra, Banking Service Chronicle
4. Analytical Reasoning By M.K. Pandey, Banking Service Chronicle

LOVELY PROFESSIONAL UNIVERSITY

Jalandhar-Delhi G.T. Road (NH-1)

Phagwara, Punjab (India)-144411

For Enquiry: +91-1824-521360

Fax.: +91-1824-506111

Email: odl@lpu.co.in